

Higher-order quantum correlations from strictly pairwise interactions: a non-associative algebraic mechanism

Xuan-Chen Liu
NPLab, Northeastern University London

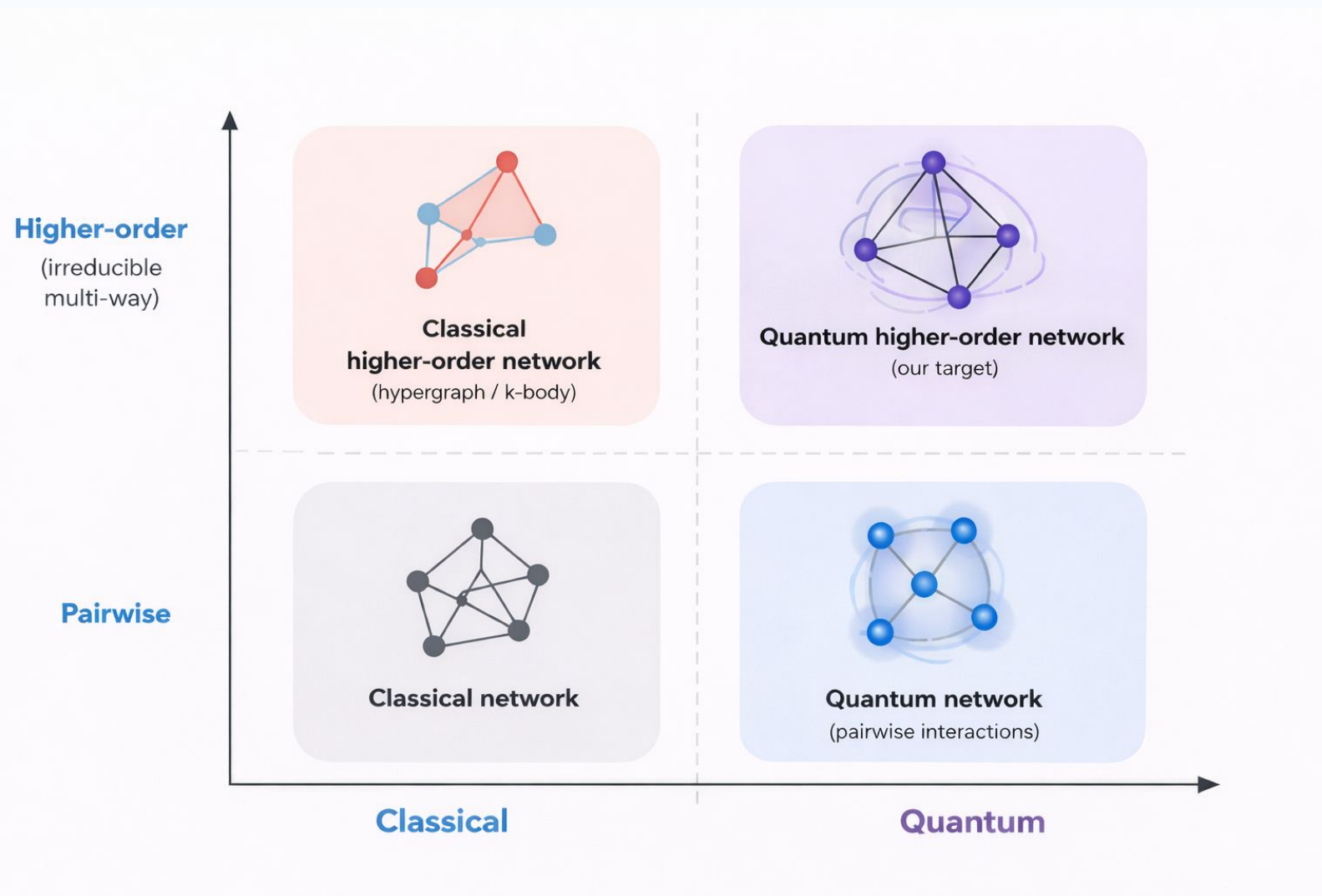


Network Science Institute
at Northeastern University



Where are we?

Method / framework



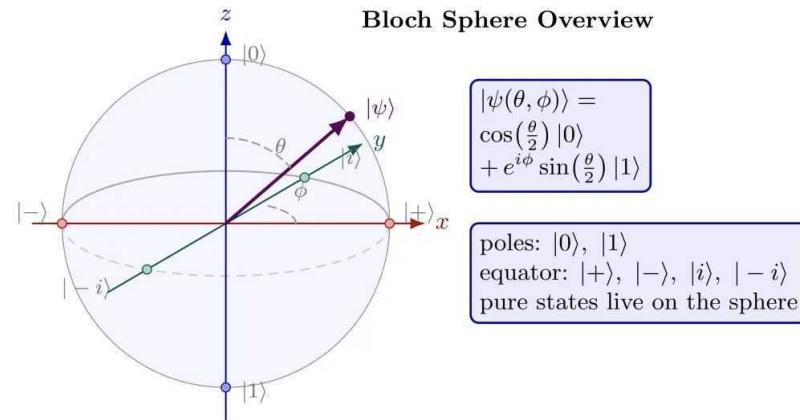
Classical vs Quantum Network Dictionary

	Classical network	Quantum network
Node	Classical variable or dynamical unit $x_i(t)$.	State of each qubit $\rho(t)$ Quantum subsystem / local Hilbert space $\mathcal{H}_i$.
Network state	A vector or collection of node variables: $\{x_i(t)\}_{i=1}^N$.	A joint quantum state $\rho(t)$ on tensor product space $\mathcal{H} = \bigotimes_i \mathcal{H}_i$
Edge	Adjacency, weight, coupling function, or influence relation A_{ij} .	Interaction Hamiltonian operator H_{ij} .
Dynamics	Example: $\frac{dx_i}{dt} = F_i(x_i) + \sum_j A_{ij} G_{ij}(x_i, x_j)$.	$\rho(t) = e^{-iHt} \rho(0) e^{iHt}$, with $H = \sum_{(ij)} H_{ij}$.

Classical networks mainly encode who interacts with whom.
Quantum networks also encode how interaction operators combine.

For network scientists

Learn some new concepts



- Full state: $\rho = \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix} = \frac{1}{2}(I + \mathbf{r} \cdot \boldsymbol{\sigma}) = \frac{1}{2}(I + r_x \sigma_x + r_y \sigma_y + r_z \sigma_z)$ A density matrix describes the quantum state of the full network, including coherence and mixedness.
- Subsystem state: $\rho_{ABC}(t) = \text{Tr}_D[\rho_{ABCD}(t)]$. Partial trace is the quantum analogue of marginalizing out unobserved degrees of freedom.
- Edge Hamiltonian: $H_{ij} = J_x^{(ij)} \sigma_x^{(i)} \sigma_x^{(j)} + J_y^{(ij)} \sigma_y^{(i)} \sigma_y^{(j)} + J_z^{(ij)} \sigma_z^{(i)} \sigma_z^{(j)}$. A quantum edge has both strength and operator direction.
- Dynamics: $\rho(t) = e^{-iHt} \rho(0) e^{iHt}$, where $H = \sum_{(ij)} H_{ij}$.
- Entropy: $S_X = -\text{Tr}[\rho_X \log_2 \rho_X]$. Information quantities are computed from reduced density matrices, not from node-level time series.
- Tripartite information (Quantum version O-information): $I_3(A:B:C) = S(A) + S(B) + S(C) - S(AB) - S(AC) - S(BC) + S(ABC)$ We use $I_3^- = \max(0, -I_3)$ as a tripartite synergy indicator.

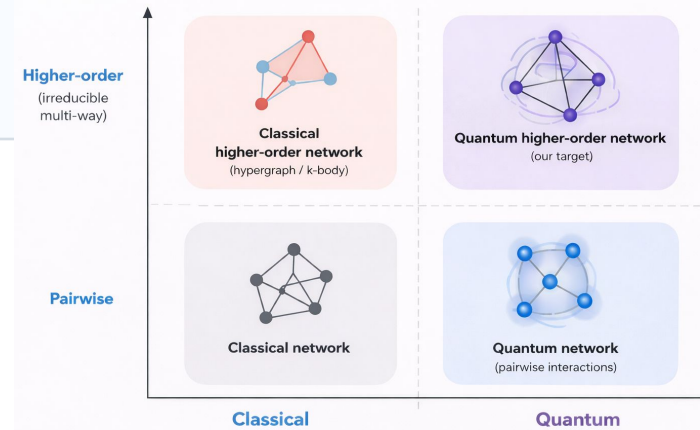
Where does the question come from?

Method / framework

Our research path

Classical Network

- > Classical Higher-order Network
- > Quantum Higher-order Network



- Classical networks capture pairwise interactions. They are used to study synchronization, spreading, robustness, controllability, collective dynamics, and structure-function relations. But now they are not enough.
→ Classical higher-order networks (hypergraphs, simplicial complexes)
- A group of nodes can interact as a unit through hyperedges or simplices. A triangle may represent not only three pairwise edges, but also a collective three-node interaction channel.
In a classical hyperedge, member nodes are at the same functional level, which are symmetric (commutative).
- In a quantum situation, interaction (combination) sequence (non-associativity) changes the outcome.

$$(H_{AB} \circ H_{BC}) \circ H_{AC} \neq H_{AB} \circ (H_{BC} \circ H_{AC})$$

Want to measure many body non-associativity? Use an **associator**.

- Question: does this algebraic structure generate irreducible higher-order quantum information?

What is the question?

Central question

Can a non-associative operator channel reveal genuinely higher-order quantum organization beyond pairwise correlations and conventional entanglement signals?

Working logic

- The Jordan associator encodes non-associative conflict among incompatible edge interactions. We use J_p and J_{tot} to describe the associator.
- The state-dependent channel J_p tells us when the state makes the non-associative channel observable.
- The state-independent (associator-dependent) channel J_{tot} tells us how much non-associative capacity the Hamiltonian contains in principle.

Project-level take-home

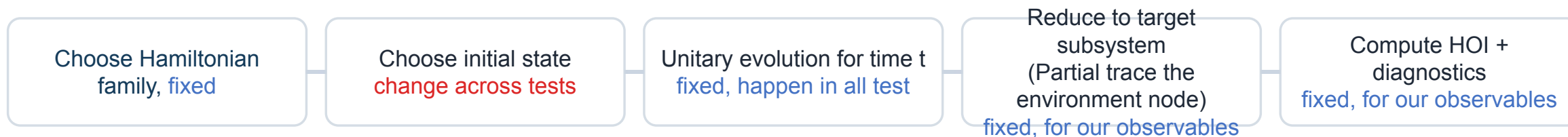
The current project interpretation is already stronger than a correlation story:

- non-associative three-edge interference acts as a generative channel for early-time tripartite synergy
- observable emergence (HOI/tripartite synergy) is controlled by a hierarchy between bare capacity and state-activated access
- richer topology changes the balance through self-averaging

Key message

The target is a mechanism: what generates HOI, when it becomes observable, and how topology changes that balance.

Numerical experiment framework



Fixed across the framework

- fixed-norm XYZ Hamiltonian family scanned on the same (θ, ϕ) grid

$$H_{ij} = J_x^{(ij)} \sigma_x^{(i)} \sigma_x^{(j)} + J_y^{(ij)} \sigma_y^{(i)} \sigma_y^{(j)} + J_z^{(ij)} \sigma_z^{(i)} \sigma_z^{(j)} \quad \vec{u}(\theta, \phi) = \begin{pmatrix} \sin\theta \cos\phi \\ \sin\theta \sin\phi \\ \cos\theta \end{pmatrix}$$

- same HOI target: I3- on the reduced face / subsystem
- same two-layer diagnostics: Jp (activated channel) and Jtot (capacity)
- same logic of early-time vs late-time interpretability

Control knobs and what they test

- **topology:** 4Q triangle $\rightarrow$ 5Q tetrahedron
- **initial state:** plus / fixed random product / Haar / random-product ensemble
- **non-associativity strength α :** exact-null ablation and recovery
- **time t:** activation-dominant vs capacity-dominant regimes
- **environment coupling and noise:** topology-sensitive robustness branches

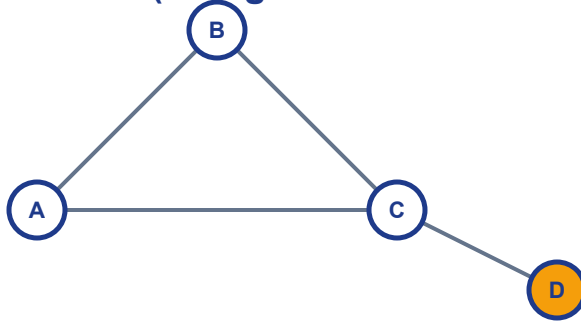
Design logic

Every test is a controlled perturbation under this framework, different parameter scan experiment can explain different perspective of mechanistic.

Systems, observables, and benchmark state choices

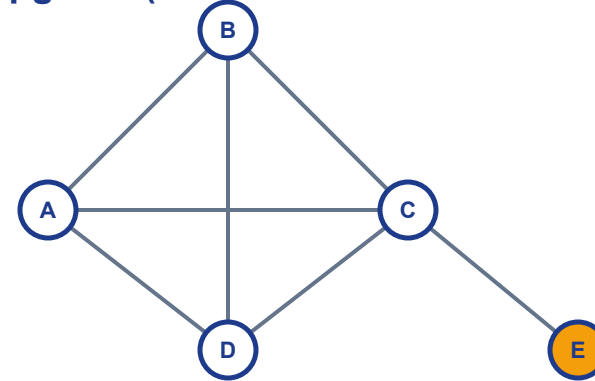
Method / framework

4Q benchmark (triangle + environment edge)



- 3 nodes, 3 edges, 1 face + environment (1 node, 1 edge)
- weak auxiliary coupling ($gD=0.2$) on the external edge
- fixed random product state = main unbiased benchmark

5Q upgrade (tetrahedron + environment node)



- 4 nodes, 6 edges, 3 faces + environment (1 node, 1 edge)
- weak/strong auxiliary coupling ($gD=0.2/1.5$) on the external edge
- four internal triangular faces are evaluated after tracing out E

Core observables used throughout the talk

Target HOI indicator: I_3^- (negative part of I_3) $I_3(A:B:C) = S(A) + S(B) + S(C) - S(AB) - S(AC) - S(BC) + S(ABC)$

State-activated diagnostic: J_p (change while dynamic evolving) | Bare-capacity diagnostic: J_{tot} (rely on associator itself)

$$m_{AB} = \text{Tr}(\rho_{ABC} A_{AB}), \quad m_{BC} = \text{Tr}(\rho_{ABC} A_{BC}), \quad m_{AC} = \text{Tr}(\rho_{ABC} A_{AC})$$

$$J_{p,\text{exp}} = \sqrt{|m_{AB}|^2 + |m_{BC}|^2 + |m_{AC}|^2}$$

$$j_{AB} = \| [H_{BC}, H_{AB}, H_{AC}]_{\circ} \|_F, \quad j_{BC} = \| [H_{AB}, H_{BC}, H_{AC}]_{\circ} \|_F, \quad j_{AC} = \| [H_{AB}, H_{AC}, H_{BC}]_{\circ} \|_F$$

$$J_{\text{tot}} = \sqrt{j_{AB}^2 + j_{BC}^2 + j_{AC}^2}$$

Benchmark step 1: Hamiltonian family & Initial state

1. Hamiltonian family $H_{ij} = J_x^{(ij)} \sigma_x^{(i)} \sigma_x^{(j)} + J_y^{(ij)} \sigma_y^{(i)} \sigma_y^{(j)} + J_z^{(ij)} \sigma_z^{(i)} \sigma_z^{(j)}$ $\vec{u}(\theta, \phi) = \begin{pmatrix} \sin\theta \cos\phi \\ \sin\theta \sin\phi \\ \cos\theta \end{pmatrix}$

total edge strength: $\sqrt{J_x^2 + J_y^2 + J_z^2} = R \quad R = \sqrt{3}$

environment edge strength: $g_D = 0.20$

2. Initial condition: Fixed random product state $|\psi_0\rangle = |\vec{n}_1\rangle \otimes |\vec{n}_2\rangle \otimes |\vec{n}_3\rangle \otimes |0\rangle$

with independently sampled: in first octant of the unit sphere $\theta \in [0.1, \pi/2] \quad \phi \in [0, \pi/2]$

scanning grid size: $20 \times 20 = 400$ points

Compare to the other settings:

- Axis-biased product state $|+++\rangle \otimes |0\rangle$: fully unentangled and highly symmetric, but biased toward axis.
- Haar-random pure state: highly entangled and locally mixed.

Benchmark step 2: Dynamics

3. Unitary evolution (the Schrödinger evolution of a closed quantum system)

In this settings:

$$\rho(t) = U(t)\rho_0 U^\dagger(t), \quad U(t) = e^{-iHt}$$

1. Closed-system coherent dynamics
2. Redistributes information across the interacting edges
3. Time t controls how far this redistribution has progressed

Why unitary evolution?

1. Minimal setting in which the Hamiltonian itself acts as the generator of structure
2. Avoids introducing noise or measurement back-action at the benchmark stage
3. Makes any generated HOI directly attributable to the internal operator algebra

Benchmark step 3: Diagnose

4. Partial trace over the auxiliary node and get the reduced triangle state

$$\rho_{ABC}(t) = \text{Tr}_D[\rho(t)]$$

Meaning of partial trace:

Sum over all possible states of D, and retain only the statistics observable on A, B, C.

This mixedness carries information about redistributed correlations and entanglement generated during the evolution.

5. Target quantity: **Tripartite Synergy Indicator** $I_3^- = \max(-I_3, 0)$ $I_3(A:B:C) = S_A + S_B + S_C - S_{AB} - S_{AC} - S_{BC} + S_{ABC}$

6. Compute associator-based diagnostics.

Jordan product: $A \circ B = \frac{1}{2}(AB + BA)$, Jordan associator: $\mathcal{A}(H_{AB}, H_{BC}, H_{CA}) = (H_{AB} \circ H_{BC}) \circ H_{CA} - H_{AB} \circ (H_{BC} \circ H_{CA})$

mediator associator: $\mathcal{A}_{BC} = (H_{AB} \circ H_{BC}) \circ H_{CA} - H_{AB} \circ (H_{BC} \circ H_{CA}) = \frac{1}{4} [H_{BC}, [H_{AB}, H_{CA}]]$

Bare algebraic capacity: $J_{\text{tot}} = \sqrt{j_{AB}^2 + j_{BC}^2 + j_{AC}^2}$ $j_{AB} = \|A_{AB}\|_F, j_{BC} = \|A_{BC}\|_F, j_{AC} = \|A_{AC}\|_F$

State-activated access: $J_{\rho, \text{exp}} = \sqrt{|m_{AB}|^2 + |m_{BC}|^2 + |m_{AC}|^2}$ $m_{AB} = \text{Tr}(\rho_{ABC} A_{AB}), m_{BC} = \text{Tr}(\rho_{ABC} A_{BC}), m_{AC} = \text{Tr}(\rho_{ABC} A_{AC})$

4Q triangle benchmark: time-resolved access → capacity crossover

Important result

Experimental setup

4Q triangle plus auxiliary edge; fixed random product initial state; same 20×20 Hamiltonian grid at every time; target observable I_{3-} on the reduced triangle.

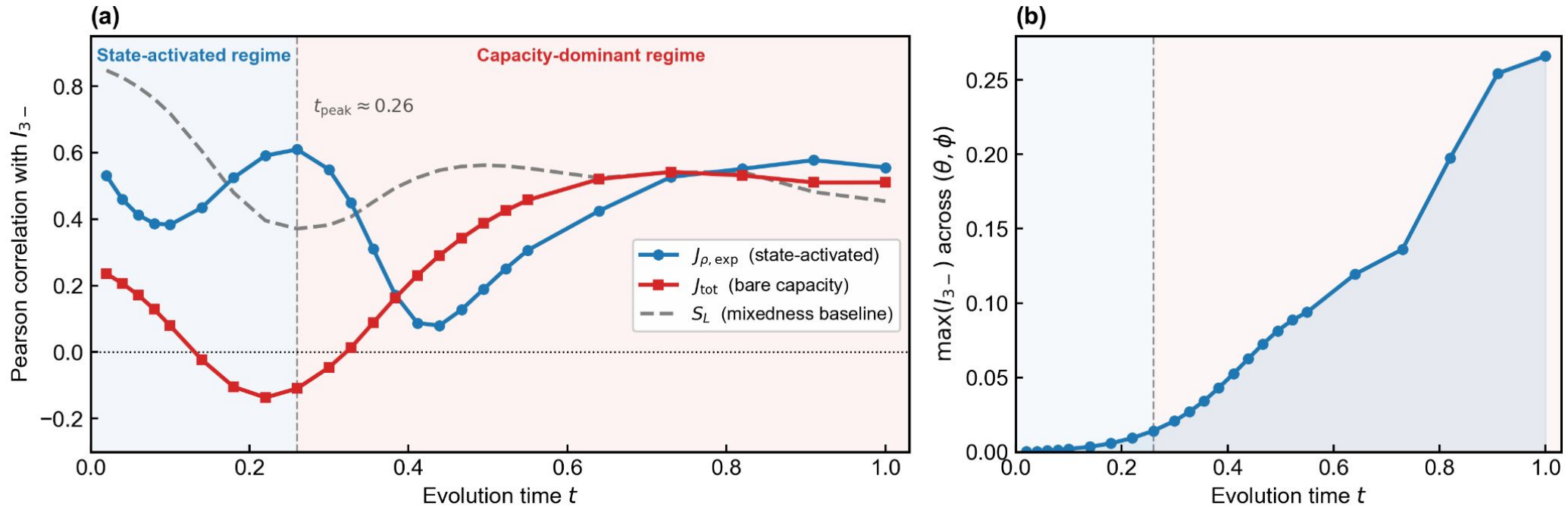
What is changed / scanned

Only evolution time t is varied. At each time slice, the full (θ, ϕ) grid is rescanned and the Pearson correlation of each diagnostic with I_{3-} is recomputed.

What the figure shows

Early times are organized by $J_{p,exp}$ (state-activated access), whereas later times are increasingly organized by J_{tot} (bare capacity).

Time-resolved associator diagnostics in the triangle network (4-qubit)
(Fixed random product initial state)



Inference from phenomenon: Early generative window is organized by $J_{p,exp}$, while later times are increasingly organized by J_{tot} .

Next question: Inside the early generative regime itself, does the HOI signal really separates from pairwise or mixedness-based surrogates at the level of individual Hamiltonian points?

Mechanism snapshot in the generative regime (4Q, fixed random product state)

Important result

Experimental setup

Same 4Q benchmark as last Slide; fixed random product initial state; a single representative early-time slice in the generative window.

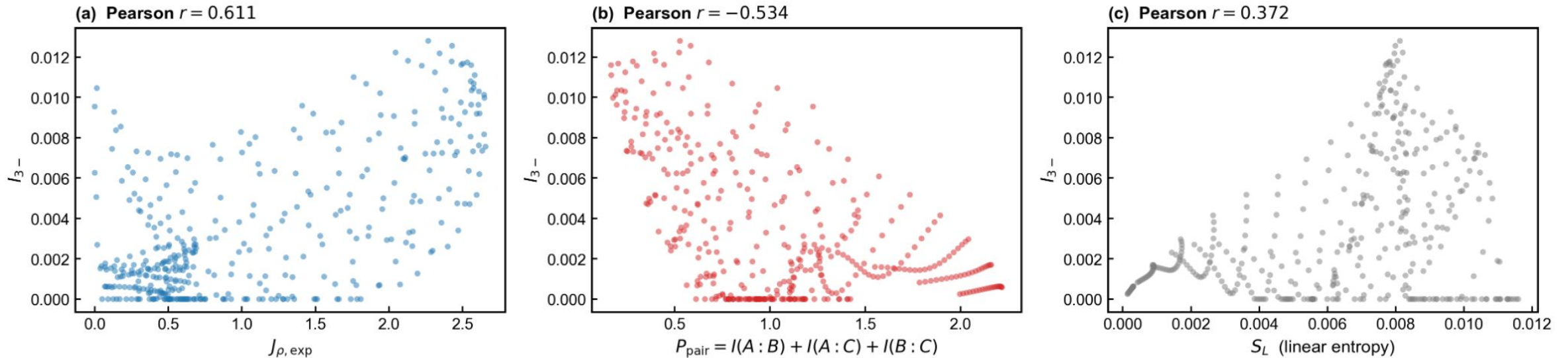
What is changed / scanned

Time is fixed ($t \approx 0.25$ in the displayed snapshot). The scanned Hamiltonian points are compared at one time across multiple candidate observables.

What the figure shows

$J_{p,\text{exp}}$ aligns positively with HOI, while the pairwise baseline is anti-aligned and mixedness is weaker: the early signal is not reducible to ordinary dyadic structure.

Mechanism snapshot at $t = 0.25$ in the generative regime
(Fixed random product initial state, $N = 400$)



Result: the early signal is not just “more pairwise correlation” in disguise; it is qualitatively tied to the state-activated non-associative channel.

Next question: how strongly this readout depends on the initial-state family, and where its applicability range begins and ends.

Boundary-condition: start from 0 entanglement vs full entanglement

Normal result

Experimental setup

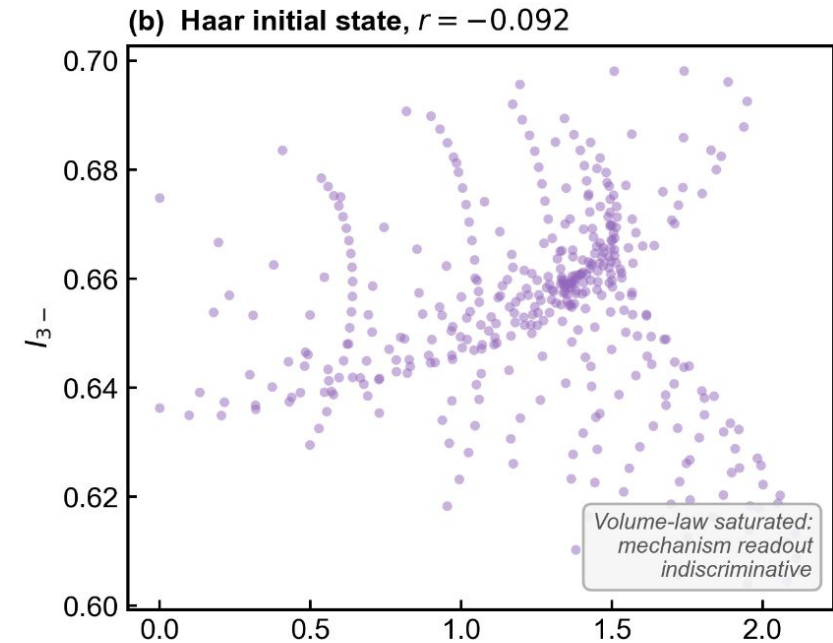
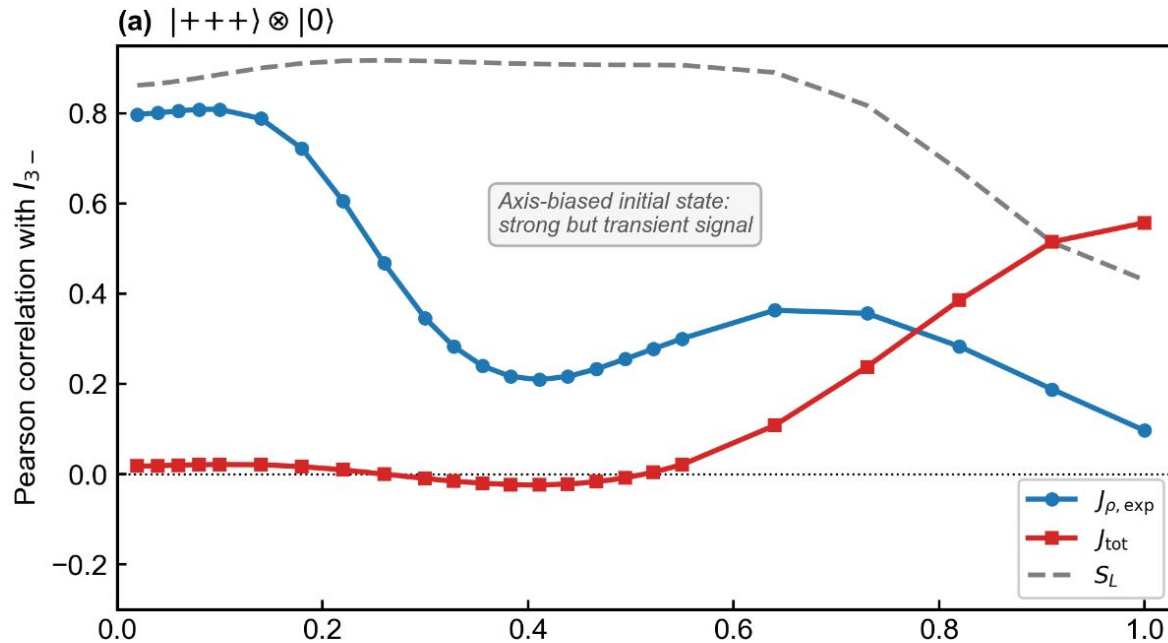
Same 4Q triangle framework and same time-resolved scan, but compare distinct initial-state families: axis-biased product state on the left and Haar-random control on the right.

What is changed / scanned

Only the initial-state family is changed. The Hamiltonian construction and diagnostics remain the same.

What the figure shows

The associator readout is strongest for clean low-entanglement starts and becomes indiscriminate for Haar-like saturated starts. This defines the applicability range of the mechanism readout.



Applicability range: it is most informative when higher-order structure is dynamically generated rather than already saturated.

Next question: We need to scan an ensemble of admissible random product states and quantify how the access channel behaves across them.

Initial-state ensemble scan: early access is state-selective; late capacity is ensemble-stable

Normal result

Experimental setup

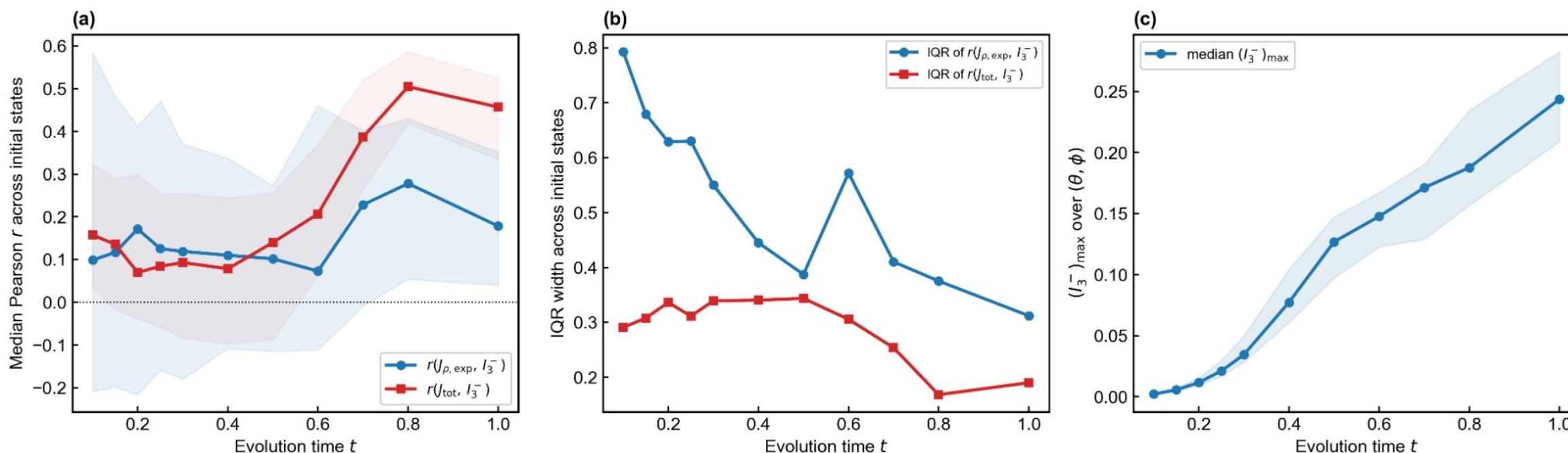
Return to the same 4Q benchmark family, but now sample a large ensemble of random product initial states and reuse the same state pool across all time slices.

What is changed / scanned

What changes is the initial state, not the Hamiltonian family. At each time, medians and IQRs are taken across the initial-state ensemble for $J_{p,exp}$, J_{tot} , and $(I_3^-)_{max}$.

What the figure shows

$J_{p,exp}$ remains far more state-sensitive, whereas J_{tot} becomes progressively more stable under ensemble averaging and behaves as a late-time structural predictor.



Message: Ensemble averaging in 4Q provides the control-side analogue of the self-averaging.

Next question: Can the same early-time mechanism predict not only HOI generation, but also where its own diagnostic channel will work well across different initial states?

Initial-state ensemble scan: the mechanism predicts its own validity domain

Normal result

Experimental setup

Uses the same random-product ensemble as last Slide. For each initial state and time slice, evaluate the within-state quality of $J_{p,exp}$ against I_3^- across the full Hamiltonian grid.

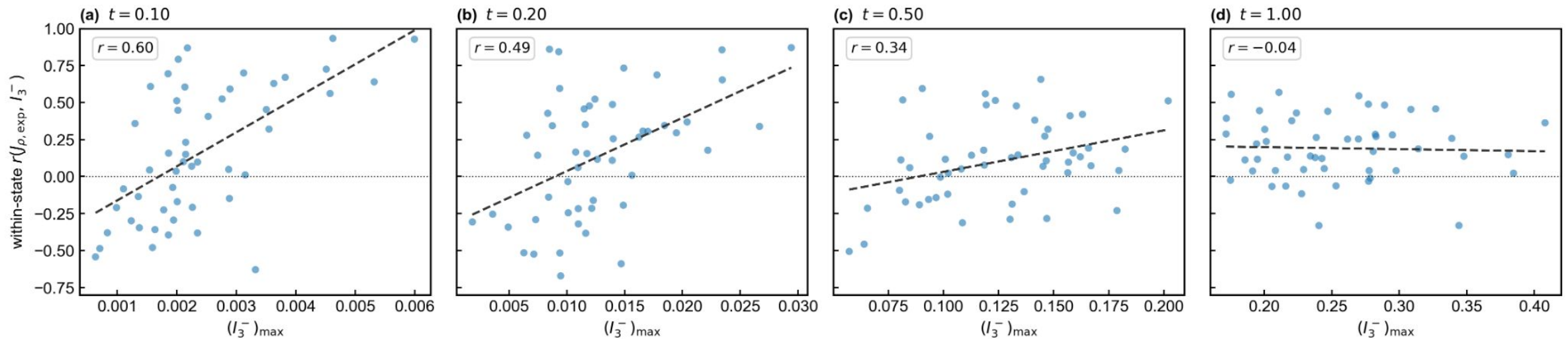
What is changed / scanned

Representative times are shown. The x-axis is the early-time HOI generation strength $(I_3^-)_{max}$; the y-axis is the within-state Pearson quality of $J_{p,exp}$.

What the figure shows

At the earliest times, states that generate stronger HOI also give higher $J_{p,exp}$ quality. The state-activated mechanism therefore predicts where its own diagnostic channel is expected to work best.

Cross-state self-consistency of the state-activated mechanism



Supporting message: State-activated mechanism predicts its own effective domain

Next question: The evidence is still showing correlation. Can we update the result?

Causal test (Exact-null ablation): remove the non-associative channel, keep pairwise dynamics

Important result

Experimental setup

$$\hat{\mathbf{n}} = \hat{\mathbf{z}} = (0,0,1)$$

$$\mathbf{J}_{ij}^{\text{null}} = \|\mathbf{J}_{ij}^{\text{ref}}\| \hat{\mathbf{n}}, \quad \hat{\mathbf{n}} = \hat{\mathbf{z}} \text{ (current implementation).}$$

$$\|\mathbf{J}_{ij}^{\text{ref}}\| = \sqrt{(J_x^{(ij),\text{ref}})^2 + (J_y^{(ij),\text{ref}})^2 + (J_z^{(ij),\text{ref}})^2}$$

$$\mathbf{J}_{ij}^{\text{null}} = (0, 0, \|\mathbf{J}_{ij}^{\text{ref}}\|) \quad H_{ij}^{\text{null}} = \|\mathbf{J}_{ij}^{\text{ref}}\| \sigma_z^i \sigma_z^j$$

What is changed / scanned

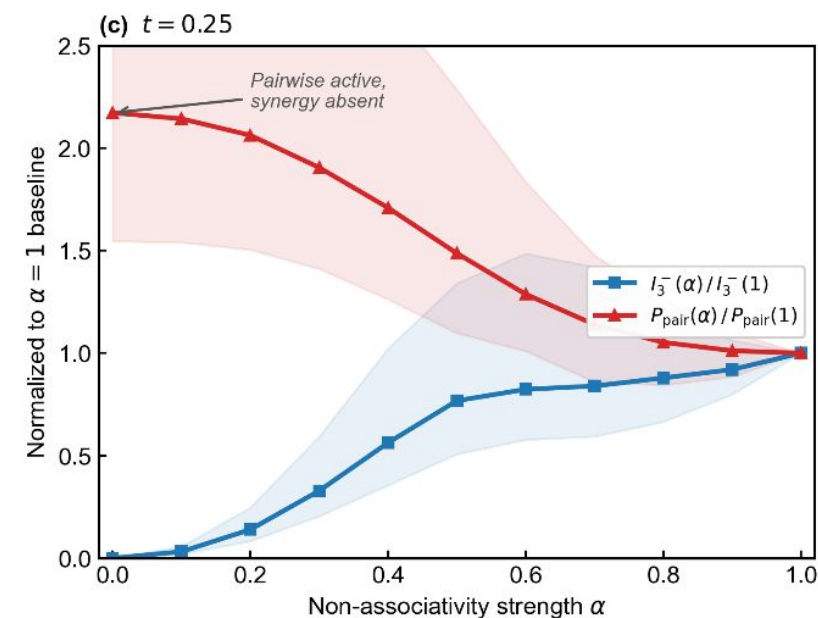
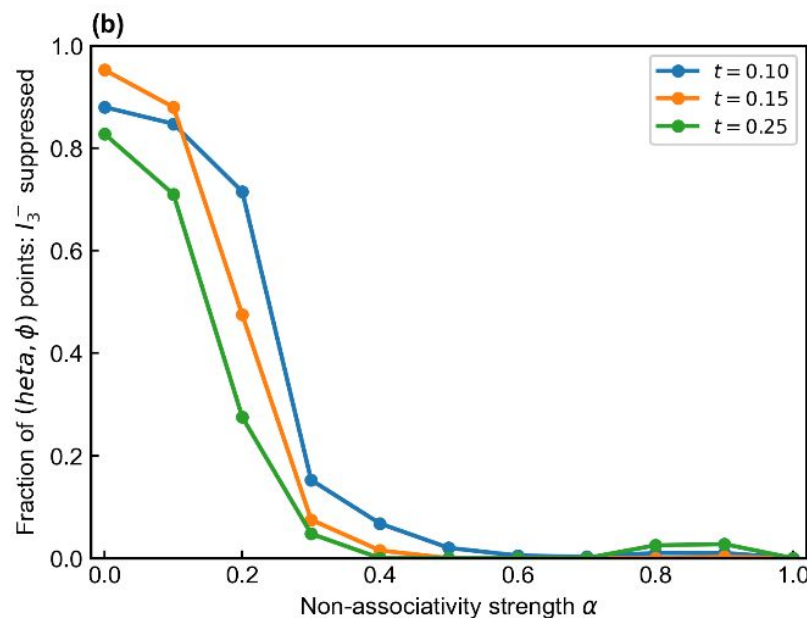
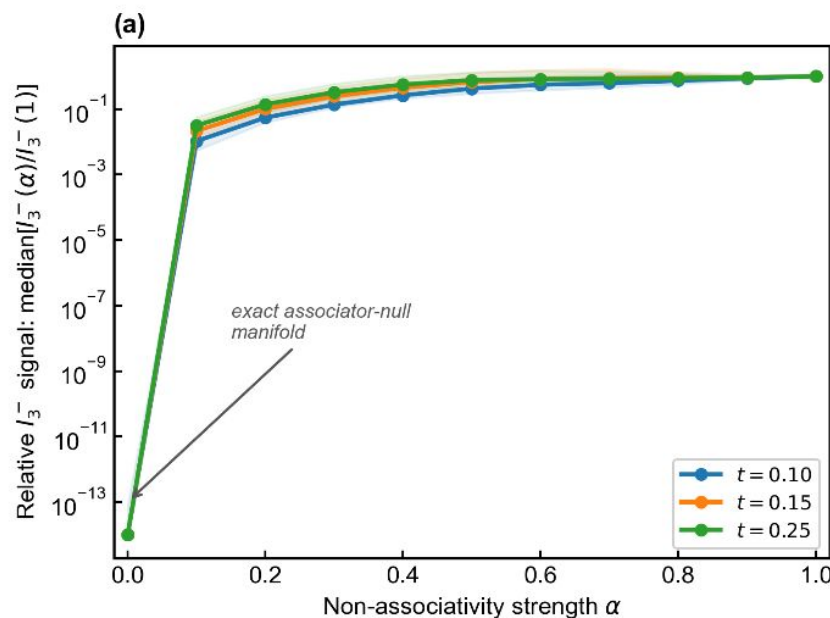
Only non-associativity strength α is varied; the scan is repeated at several early times (shown: $t = 0.10, 0.15, 0.25$) over the same 20×20 Hamiltonian grid.

What the figure shows

At $\alpha = 0$ the HOI signal collapses to the machine floor while pairwise dynamics remains active. This is a counterfactual necessity test, not just another correlation plot.

$$\mathbf{J}_{ij}^{\text{mix}}(\alpha) = (1 - \alpha) \mathbf{J}_{ij}^{\text{null}} + \alpha \mathbf{J}_{ij}^{\text{ref}}, \quad \mathbf{J}_{ij}^{(\alpha)} = \|\mathbf{J}_{ij}^{\text{ref}}\| \frac{\tilde{\mathbf{J}}_{ij}(\alpha)}{\|\tilde{\mathbf{J}}_{ij}(\alpha)\|}$$

Causal test: interpolation from the associator-null manifold



Causal test The early-time HOI signal collapses to machine floor, while pairwise mutual-information and pairwise-negativity backgrounds remain active.

Next question: When we restore non-associativity continuously, any other “dynamic” information?

Causal test: Two-layer recovery from the null manifold: capacity rises first, HOI follows later

Important result

Experimental setup

Uses the same α -interpolation data as Slide 7; all curves are normalized to their $\alpha = 1$ values to compare recovery on a common scale.

What is changed / scanned

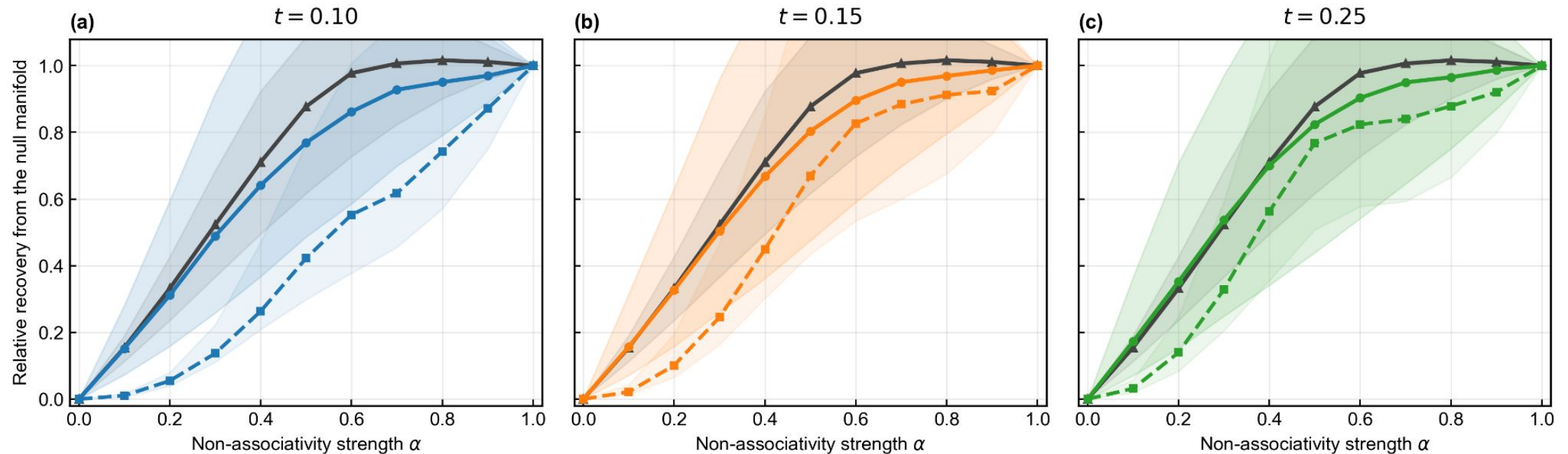
α is scanned from the exact null manifold back to the original family, while three channels are tracked simultaneously: J_{tot} , $J_{\rho, \text{exp}}$, and I_{3-} .

What the figure shows

J_{tot} recovers earliest, I_{3-} latest, and $J_{\rho, \text{exp}}$ in between. Bare capacity is necessary but not sufficient: state activation is the practical bottleneck.

Two-layer recovery from the associator-null manifold

$J_{\text{tot}}(\alpha)/J_{\text{tot}}(1)$ $J_{\rho, \text{exp}}(\alpha)/J_{\rho, \text{exp}}(1)$ $I_{3-}(\alpha)/I_{3-}(1)$



Hierarchical bottleneck

Restoring structure is easier than restoring observable higher-order signal.

Next question: Crossover, necessity, and hierarchy are detected in the closed system. Now we step further to robustness.

4Q open-system branch: higher-order structure stays robust to noise

Important result

Experimental setup

Prepared top-k 4Q families are evolved unitarily up to t_{prep} and then subjected to noise on the tetrahedron nodes only; the environment node remains coherent.

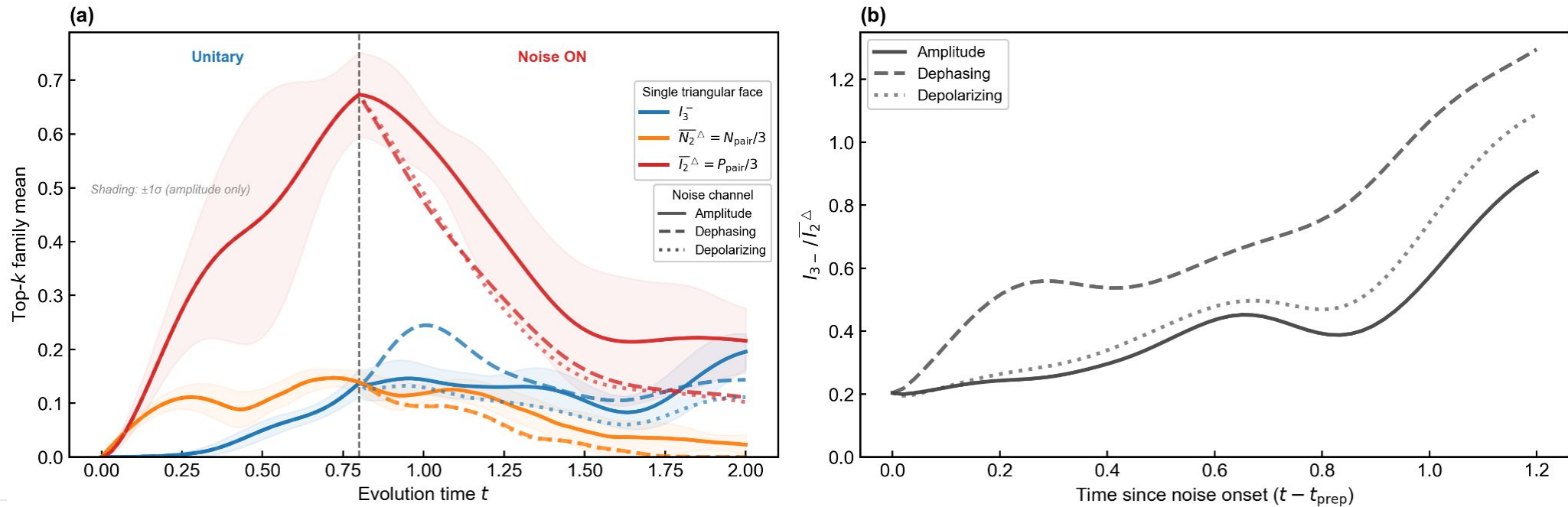
What is changed / scanned

What changes is the noise channel (amplitude / dephasing / depolarizing) and time since noise onset. Internal HOI is compared against baselines and edge observables.

What the figure shows

Higher-order structure remains comparatively robust, but the environment edge does not provide a faithful proxy for the internal HOI dynamics.

Noise robustness in the 4-qubit triangle extension
(top-k prepared families, $g_D = 0.2$, $t_{\text{prep}} = 0.8$, $\gamma = 0.2$, $k = 12$)



message: Tripartite synergy survived after introducing noise. Pairwise negativity (entanglement) died while only classic pairwise mutual information remains (not useful).

Next question: 4-qubit closed-system and open-system are all tested. Let's go to bigger system.

5Q tetrahedral extension: same framework, different balance between access and capacity

Important result

Experimental setup

Upgrade the 4Q triangle to a 5Q tetrahedron plus environment node; fixed random product initial state; at each time, trace out E and evaluate all four triangular faces.

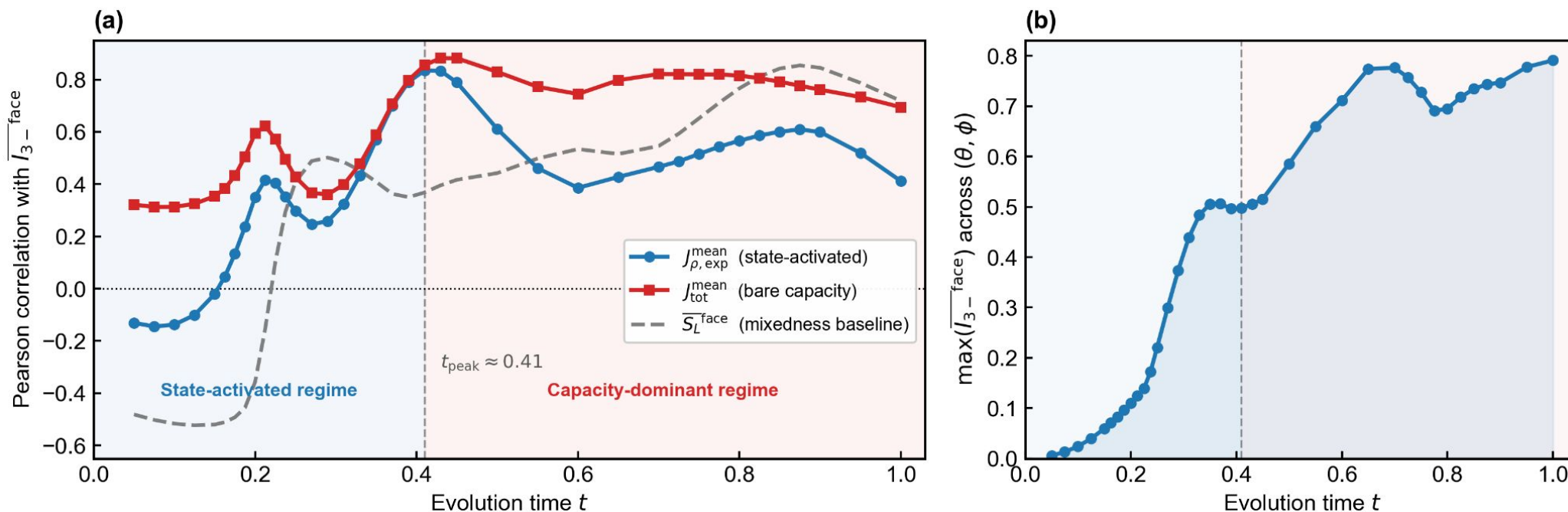
What is changed / scanned

Only topology is changed relative to the 4Q benchmark. Time is rescanned exactly as before, but the diagnostics are now face-averaged over a richer simplicial structure.

What the figure shows

Bare capacity becomes informative much earlier, while state activation still has a peak but behaves more like a correction on top of a broader structural reservoir.

Time-resolved associator diagnostics in the tetrahedral network (5-qubit)
(Fixed random product initial state, $g_E = 0.2$)



Topology upgrade The 5Q story is not "4Q again": self-averaging shifts predictive weight toward the bare-capacity channel.

Next question: We have several triangle faces, does the external coupling affects all faces uniformly?

5Q environment coupling: 4 faces in tetrahedron exhibit different reaction to the environment

Normal result

Experimental setup

Choose one representative high-signal 5Q tetrahedral point (shown in the figure title), keep the internal Hamiltonian point fixed, and trace how face-resolved observables respond to the external coupling.

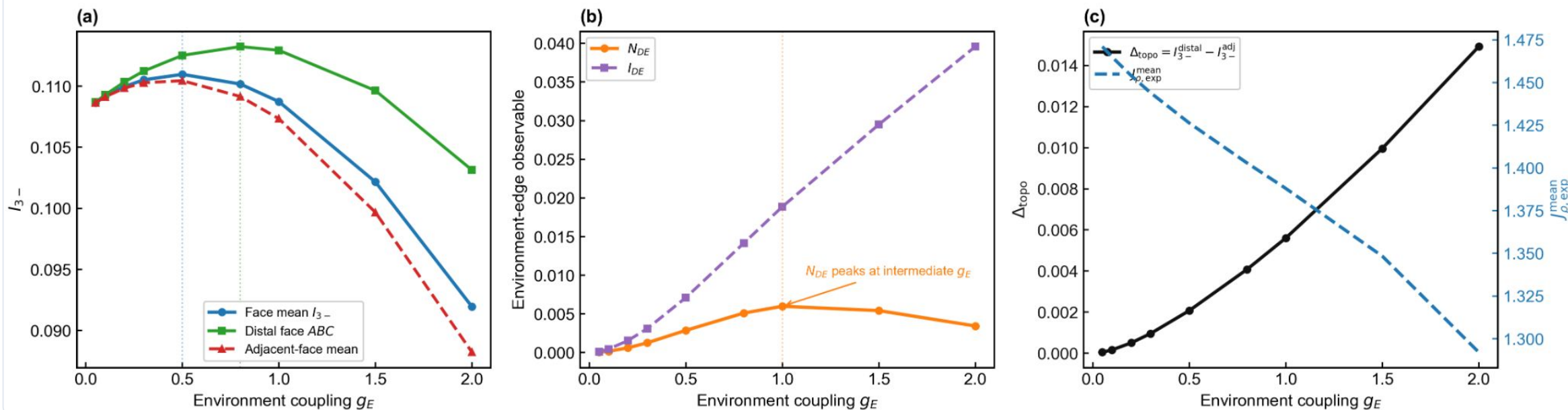
What is changed / scanned

Only environment coupling g_E is scanned. The displayed curves compare face mean, distal face, adjacent-face mean, and environment-edge observables.

What the figure shows

The distal face remains protected even after the face mean starts to turn over; the environment induces genuine topology-sensitive differentiation rather than a uniform renormalization.

Figure 5: Environment-coupling scan on the best 5-qubit tetrahedral point
($\theta = 0.501$, $\phi = 0.857$, $t = 0.20$)



message

This is the network-level version of the mechanism story: where the environment attaches matters for how higher-order organization is redistributed.

Next question: Is this richer higher-order organization is also robust under open-system noise?

5Q open-system branch: higher-order structure stays robust to noise

Experimental setup

Prepared top-k 5Q families are evolved unitarily up to t_{prep} and then subjected to noise on the tetrahedron nodes only; the environment node remains coherent.

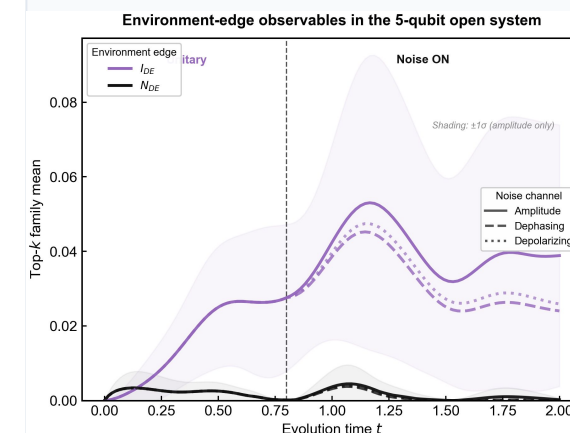
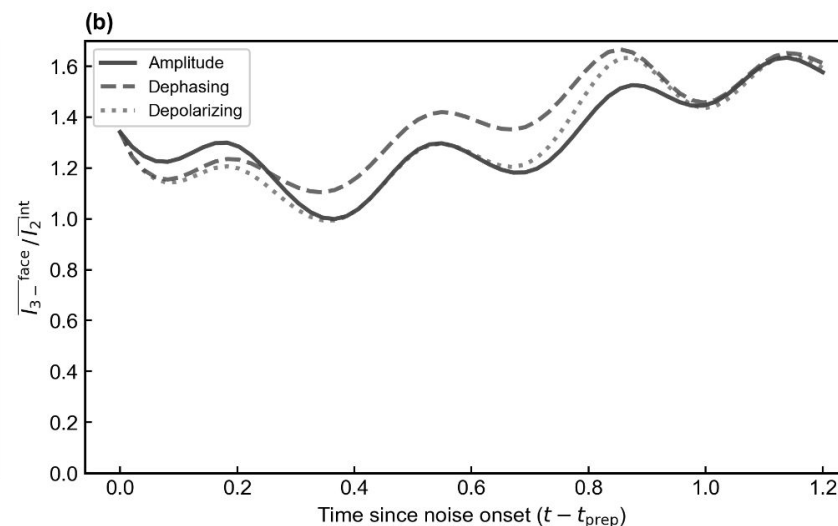
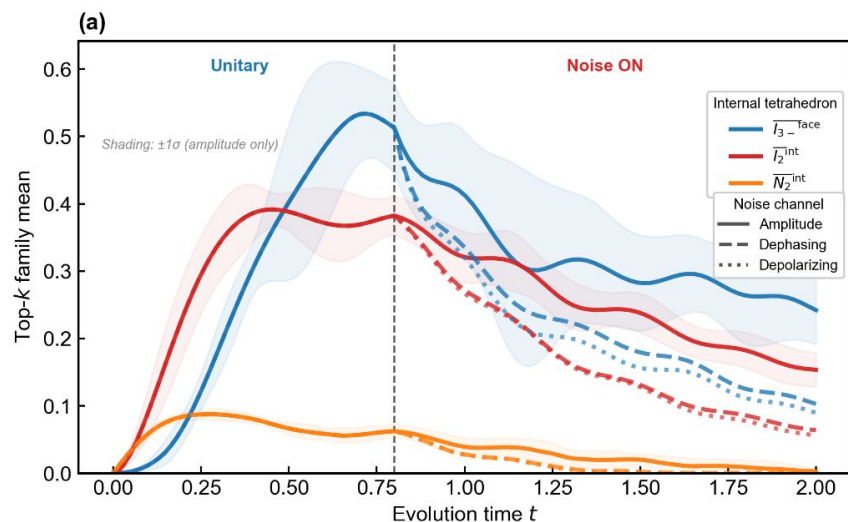
What is changed / scanned

What changes is the noise channel (amplitude / dephasing / depolarizing) and time since noise onset. Internal HOI is compared against baselines and edge observables.

What the figure shows

Higher-order structure remains comparatively robust, but the environment edge does not provide a faithful proxy for the internal HOI dynamics.

Noise robustness in the 5-qubit tetrahedral extension
(top-k prepared families, $g_E = 0.2$, $t_{\text{prep}} = 0.8$, $\gamma = 0.2$, $k = 10$)



Inset: environment-edge observables

The internal organization is not reducible to what happens on the external edge.

Supporting message

This reinforces the 5Q interpretation: the internal higher-order organization is not reducible to what happens on the external edge.

Next question: Larger systems.

Transverse-field Ising model: results from a standard model

Edge hamiltonian $H_{ij}^{\text{TF}} = J_{ij}Z_iZ_j + \frac{h}{2}(X_i + X_j)$ $H_{\Delta}^{\text{TF}} = J_{AB}Z_AZ_B + J_{BC}Z_BZ_C + J_{AC}Z_AZ_C + h(X_A + X_B + X_C)$

total edge strength: $J_{\text{tot}}^2 = h^4 \sum_{(ij) \in \{AB, BC, AC\}} J_{ij}^2$ $J_{AB}^2 + J_{BC}^2 + J_{AC}^2 = R^2$

bare capacity becomes $J_{\text{tot}} = h^2 R$

Parametrize edge magnitudes: $K_{AB} = R \sin \theta \cos \phi$, $K_{BC} = R \sin \theta \sin \phi$, $K_{AC} = R \cos \theta$,

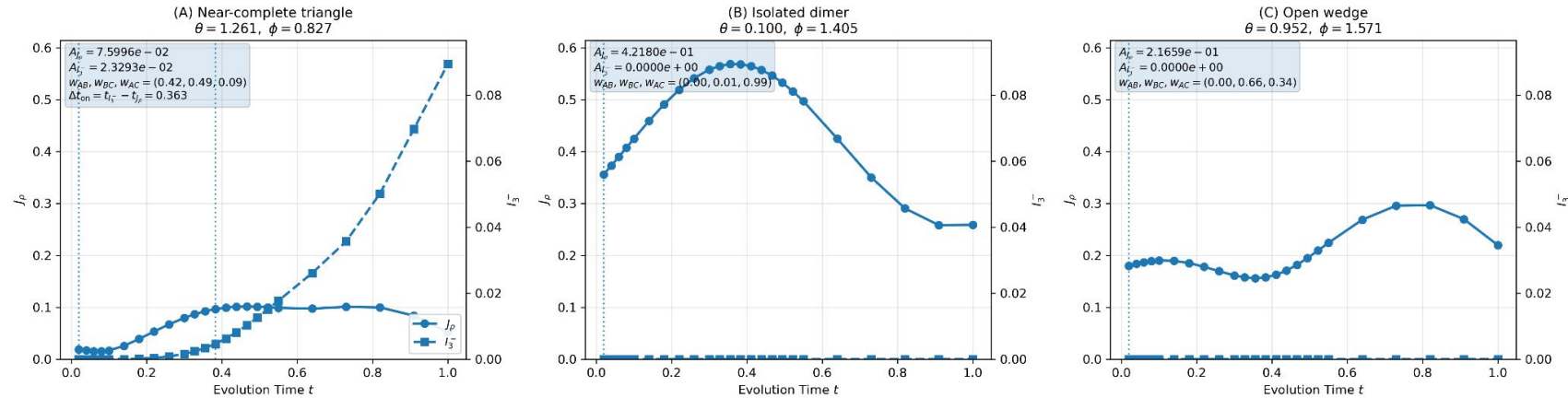
$$K_{AB}^2 + K_{BC}^2 + K_{AC}^2 = R^2 \quad \theta \in [0, \pi/2], \quad \phi \in [0, \pi/2].$$

every point in the scan has the same bare capacity $J_{\text{tot}} = h^2 R$.

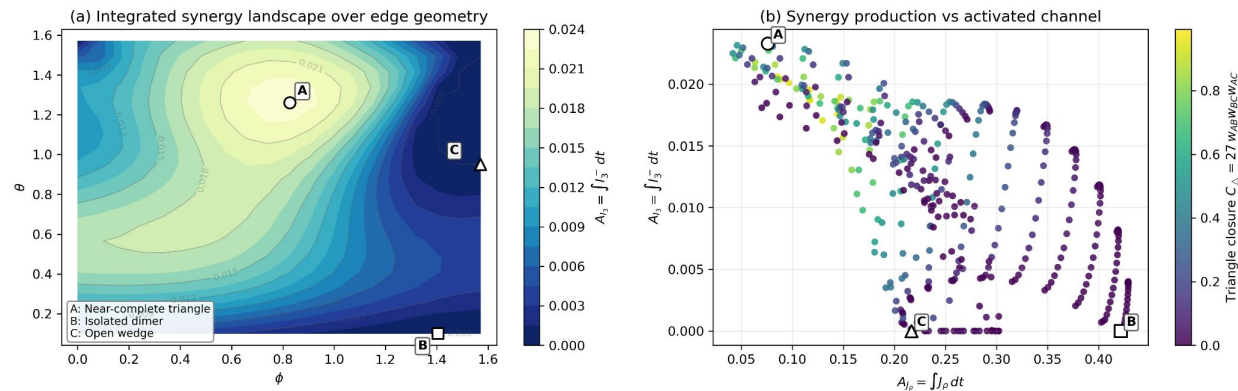
Transverse-field Ising model: results from a standard model

Important extra result

Representative TF-Ising trajectory examples
FM_near_critical_like | $\lambda = h/\bar{J} = 1.00$ | ferromagnetic-like, near-critical-like finite-size regime



Geometry dependence of tripartite synergy indicator in the TF-Ising triangle
FM_near_critical_like | $\lambda = h/\bar{J} = 1.00$ | ferromagnetic-like, near-critical-like finite-size regime



$$w_{ij} = \frac{K_{ij}^2}{K_{AB}^2 + K_{BC}^2 + K_{AC}^2}$$

$$C_{\Delta} = 27 w_{AB} w_{BC} w_{AC}$$

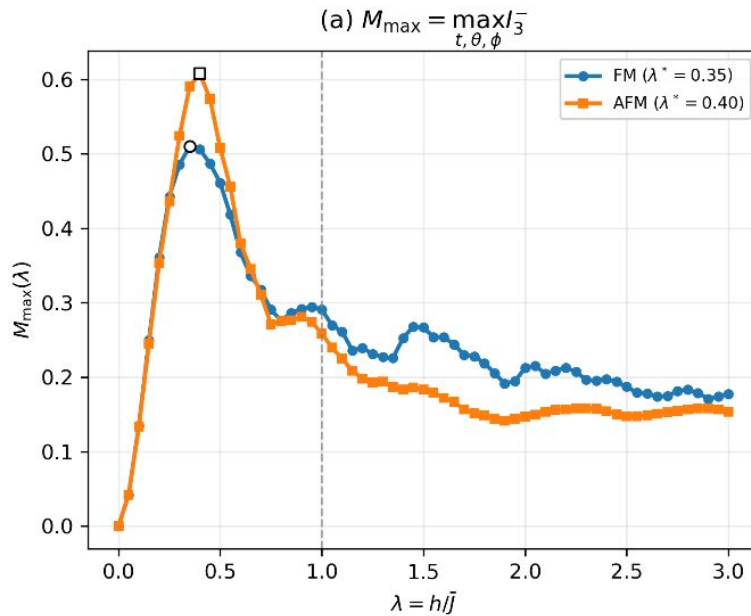
Topology dependent I_3^- generation still needs J_{ρ} , but also relies on the topology of the triangle.

Next question: How will different h influence the results.

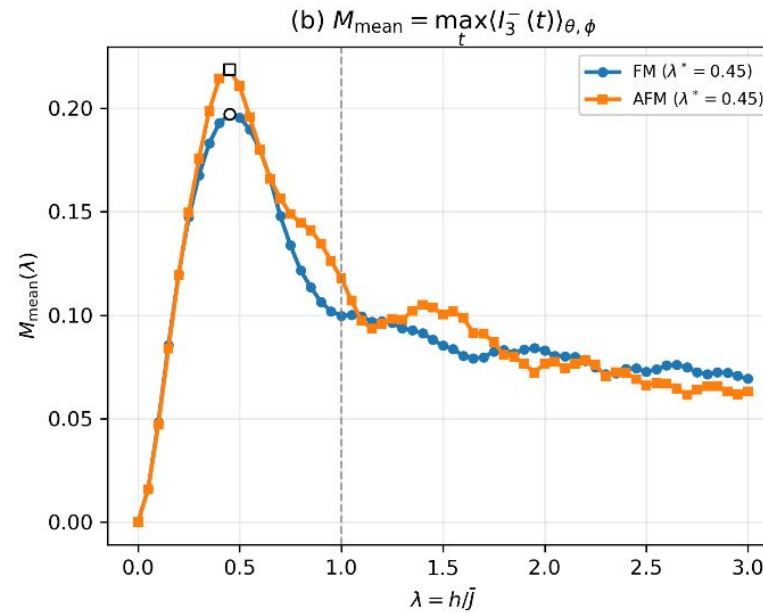
Transverse-field Ising model: results from a standard model

Important extra result

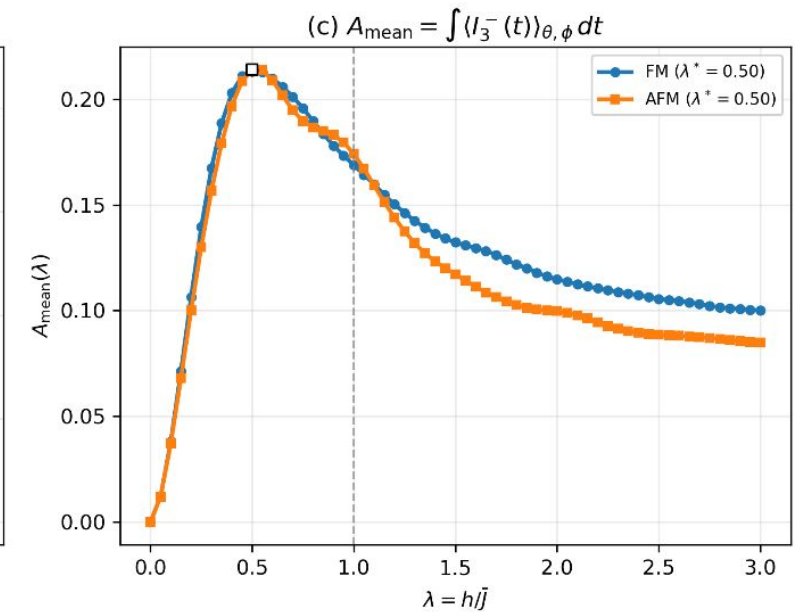
Lambda scan in the TF-Ising triangle



$$M_{\max}(\lambda) = \max_{t, \theta, \phi} I_3^-(t; \theta, \phi),$$



$$M_{\text{mean}}(\lambda) = \max_t \langle I_3^-(t; \theta, \phi) \rangle_{\theta, \phi},$$



$$A_{\text{mean}}(\lambda) = \int \langle I_3^-(t; \theta, \phi) \rangle_{\theta, \phi} dt.$$

Finite-size effect? Maybe all peaks go left because the experiments were taken in a small system.

Next question: Analytical analysis and numerical experiments for larger system.

Results we have

What each parameter experiment reveals

Time t

activation-dominant $\rightarrow$ capacity-dominant crossover

Null interpolation α

necessity of the non-associative channel + hierarchical recovery

Initial state

access gating and endogenous validity domain of $J_{p,exp}$

Topology (4Q $\rightarrow$ 5Q)

self-averaging shifts predictive weight toward bare capacity

Environment / noise

topology-sensitive differentiation branches

Message take-home

For quantum information

We identify a concrete mechanism for generating quantum HOI.

- Irreducible tripartite quantum correlations are generated through a non-associative three-edge channel.
- Their observable emergence is controlled by a two-layer hierarchy: state-activated access and bare algebraic capacity
- Self-averaging makes the bare-capacity channel informative earlier.

For network science

If we want to uncover the mechanism of HOI in networked systems, try to answer: who plays the role of “non-commutativity”? Several possible ways:

- Nonlinear dynamics: Mode coupling in nonlinear dynamics can generate irreducible higher-order effects beyond linear diffusion.
- Non-decomposable hyperedge weights: If a 3-body hyperedge weight cannot be reduced to pairwise weights, it may play the same role as a nonzero associator.
- Information-geometric curvature: Non-flat information geometry signals that joint structure cannot be reconstructed from pairwise marginals alone.

Final message

The project is a mechanism story, with necessity, hierarchy, and topology-dependent redistribution all tested within one framework.

Thanks for listening!